

Full Name: _____

Group: _____

There is no penalty for wrong answers. Do not leave questions blank. No outside materials or devices are allowed.

Item	Sec.	Points	Your Answer	Item	Sec.	Points	Your Answer
1	A	0.5	A	16	A	0.5	A
2	A	0.5	B	17	A	0.5	B
3	A	0.5	C	18	A	0.5	A
4	A	0.5	B	19	A	0.5	A
5	A	0.5	C	20	A	0.5	B
6	A	0.5	B	21	A	0.5	B
7	A	0.5	C	22	A	0.5	B
8	A	0.5	D	23	A	0.5	C
9	A	0.5	C	24	A	0.5	B
10	A	0.5	C	25	B	1.0	-
11	A	0.5	B	26	B	1.0	-
12	A	0.5	B	27	B	1.0	-
13	A	0.5	C	28	C	1.5	-
14	A	0.5	C	29	C	1.5	-
15	A	0.5	B	30	D	2.0	-
Subtotal A: 12.0				Subtotal B–D: 8.0			
Total: 30 items = 20.0 points							

Part A — Test Questions ($24 \times 0.5 = 12$ points)

1. A ViT trained on a small dataset performs worse than a similarly sized CNN. Which explanation is most plausible?
A. ViTs have weaker spatial priors
B. Transformers cannot read RGB channels directly
C. CNNs cannot model long-range relations
D. CNNs automatically see the whole image
2. Two augmentations of the same photograph are encoded by a contrastive model. During training the objective encourages:
A. large distance between the two views
B. similar embeddings for the two views
C. identical embeddings for all images
D. orthogonal embeddings for every pair
3. A CLIP-like model receives the prompts “a dog”, “a wolf”, and “a fox”. What determines the predicted class for an image?
A. the prompt with the highest token score
B. the most frequent class in pretraining
C. the closest text embedding
D. the prompt with the longest caption
4. Why can a CNN detect the same visual pattern in different parts of an image?
A. each pixel has its own independent classifier
B. the same filter is reused across locations
C. the network removes spatial structure at the input
D. every layer sees only the image center
5. Why is object detection usually harder than image classification?
A. detectors cannot use pretrained backbones
B. detectors only work on grayscale images
C. the model must find where objects are and what they are
D. the task never allows data augmentation
6. Which scenario most strongly indicates dataset shift?
A. train and validation scores stay close
B. performance drops on a new camera
C. training loss decreases each epoch
D. two augmentations get similar outputs
7. A defect dataset contains 95% normal and 5% defective images. A classifier predicts “normal” for every image and gets 95% accuracy. Which conclusion is most appropriate?

- A. the model is already strong from accuracy alone
B. precision alone confirms good defect detection
C. accuracy is misleading; check recall or F1
D. class balance here guarantees good calibration
8. A transformer processes image patches as tokens. Which property of attention enables global context modeling?
A. each token sees only adjacent tokens
B. tokens interact through fixed kernels
C. tokens interact only inside windows
D. any token can attend to any token
9. Why can CLIP perform classification on categories unseen during training?
A. its classifier head stores every class
B. it retrains itself during inference
C. image embeddings are compared with text prompts
D. it always generates a caption first
10. Why are VAE reconstructions often blurrier than outputs from strong diffusion models?
A. VAEs cannot use latent variables
B. VAEs cannot condition on an input
C. the decoder averages plausible outputs
D. diffusion models never add noise
11. Two object detectors are evaluated on the same validation set. Recall that:
- **mAP@0.5** evaluates detection when the predicted box overlaps the ground truth with $\text{IoU} \geq 0.5$.
 - **mAP@[0.5:0.95]** averages AP across stricter IoU thresholds up to 0.95.

Detector	mAP@0.5	mAP@[0.5:0.95]
A	0.82	0.44
B	0.79	0.60

Which detector localizes objects more precisely?

- A. Detector A
B. Detector B
C. Both detectors have similar localization quality
D. The metrics do not allow comparison
12. Which statement best distinguishes semantic and instance segmentation?
A. instance segmentation predicts captions
B. it separates same-class objects

- C. semantic segmentation estimates depth
 - D. semantic segmentation returns only boxes
13. A pretrained SSL representation is fine-tuned on a small labeled dataset. Which effect is most likely?
- A. slower convergence from larger features
 - B. more labels become necessary to train
 - C. better performance with fewer labels**
 - D. the model loses its ability to generalize
14. In transformer-based perception models (e.g., detection or VLMs), cross-attention is typically used to:
- A. increase the spatial resolution of feature maps
 - B. remove redundant image tokens
 - C. condition one representation on another**
 - D. eliminate the need for positional encoding
15. When adapting a large VLM using LoRA, the main motivation is:
- A. increase the input image resolution
 - B. update only a few weights**
 - C. replace self-attention with convolutions
 - D. remove the text encoder entirely
16. A SAM-like model receives one positive point exactly between two touching tools. What is most likely?
- A. the prompt is ambiguous, so either mask may appear**
 - B. the model must segment the larger tool only
 - C. the model must output both masks at once
 - D. the point is ignored near touching objects
17. Why do monocular depth estimation systems often predict depth only up to an unknown scale?
- A. the neural network cannot learn metric units
 - B. multiple scaled 3D scenes can project to the same image**
 - C. RGB cameras cannot detect object boundaries
 - D. the training dataset contains too few images
18. Why are heatmap-based keypoint detectors often preferred to direct coordinate regression under occlusion?
- A. heatmaps can represent location uncertainty**
 - B. heatmaps remove the need for labels
 - C. heatmaps directly recover full 3D pose
 - D. heatmaps guarantee visible joints

19. In contrastive learning with a batch of images, negative samples are typically:
- A. other images in the batch**
 - B. extra views of the same image
 - C. random Gaussian noise vectors
 - D. segmentation masks from the dataset
20. Why do deeper CNN layers capture more global image structure?
- A. deeper layers always use larger kernels
 - B. the receptive field grows across layers**
 - C. deeper layers remove spatial information
 - D. deeper layers operate only on RGB values
21. Why is a separate validation dataset used when training vision models?
- A. to increase training speed
 - B. to estimate how well the model generalizes**
 - C. to remove mislabeled images
 - D. to compute convolution kernels
22. Which training signal aligns image and text embeddings in CLIP-style models?
- A. standard supervised classification loss
 - B. a contrastive similarity loss**
 - C. pixel-wise reconstruction loss
 - D. reinforcement learning from rewards
23. Which change most directly improves robustness to lighting variation in deployment?
- A. remove normalization from the backbone
 - B. increase the depth of the network
 - C. use photometric augmentation**
 - D. replace CNN layers by MLP blocks
24. In early YOLO models, the image is divided into a grid. What does each grid cell predict?
- A. only the class label of the image
 - B. bounding boxes and class probabilities**
 - C. segmentation masks for all objects
 - D. camera calibration parameters

Part B — Basic Practical ($3 \times 1 = 3$ points)

25. A student must build a classifier for fine-grained industrial defects using only 3,000 labeled images. They consider three starting points:

1. a ViT trained from scratch,
2. a pretrained ResNet50 fine-tuned on the dataset,
3. a CLIP-like model used zero-shot with generic prompts.

Tasks.

1. Choose the **best starting point**.
2. Give **two reasons** for your choice.

Solution: Best choice: **a pretrained ResNet50 fine-tuned on the dataset.**

Possible reasons:

- With only 3,000 labeled images, a pretrained CNN is usually more data-efficient than a ViT trained from scratch.
- CNNs have stronger spatial inductive bias and often work well on smaller vision datasets.
- Fine-tuning a pretrained ResNet50 is a reliable baseline before trying heavier or more specialized models.
- Zero-shot CLIP with generic prompts is unlikely to separate subtle, fine-grained industrial defect classes well enough.

26. You need a classifier for a mobile robot. The requirements are:

- inference speed at least 30 FPS,
- validation accuracy at least 82%,
- deployment on limited hardware.

You test four models:

Model	Val. Acc.	Latency / image	Params
MobileNetV3	79%	8 ms	5M
ResNet50	84%	22 ms	26M
ViT-B/16	85%	48 ms	86M
CLIP ViT-B (zero-shot)	69%	60 ms	86M

Tasks. Which model is the most reasonable choice? Give **two** reasons.

Solution: Best choice: **ResNet50**.

Possible reasons:

- It satisfies the accuracy requirement (84%).
- It satisfies the speed requirement: 22 ms per image is about 45 FPS.
- It is much lighter and faster than ViT-B/16 and CLIP ViT-B while remaining more accurate than MobileNetV3.

27. A CLIP-like model is used for zero-shot classification of three surgical instruments:

- scissors
- forceps
- clamp

A student proposes the prompt set:

- “instrument”
- “surgical tool”
- “metal device”

Tasks.

1. Explain why this prompt design is weak.
2. Propose one **better class-specific prompt strategy**.
3. Propose one method to reduce **prompt sensitivity** without full retraining.

Solution: Why weak:

- The prompts are too generic and do not correspond cleanly to the actual target classes.
- The text embeddings for these prompts are likely to be close to each other, so visually similar instruments remain hard to separate.

Possible improvements:

- Use class-specific prompts such as “a photo of surgical scissors”, “a photo of surgical forceps”, and “a photo of a surgical clamp”.
- Use multiple templates per class and average the resulting text embeddings.

Part C — Advanced Practical (1.5 + 1.5 = 3.0 points)

28. A student sketches the following segmentation architecture for small objects:

Input image $256 \times 256 \rightarrow$ Encoder $\rightarrow$ Bottleneck $\rightarrow$ Decoder $\rightarrow$ Mask 64×64

Additional notes:

- There are **no skip connections** between encoder and decoder.
- The final mask is compared with the ground truth after simply resizing it.

Tasks.

1. Identify **two** architectural issues.
2. Describe one likely failure mode on small or thin objects.
3. Propose **two** improvements.

Solution: Possible issues:

- **No skip connections:** the decoder receives weak high-resolution spatial detail, which hurts localization and boundaries.

- **Low-resolution output mask:** predicting only 64×64 and resizing it back can blur boundaries and erase small objects.

Likely failure mode:

- Small or thin objects may disappear, merge into the background, or have highly inaccurate borders.

Possible improvements:

- Add U-Net-style skip connections.
- Predict masks at higher resolution, or upsample progressively with dense supervision.
- Use a decoder designed for dense prediction instead of relying only on a coarse bottleneck representation.

29. A self-supervised representation learning system uses the objective

$$L = \|z^{(1)} - z^{(2)}\|_2^2,$$

where $z^{(1)}$ and $z^{(2)}$ are embeddings of two augmentations of the same image.

After training, almost all images produce nearly identical embeddings.

Tasks.

1. Show that assigning the same constant embedding to every image is a global minimum of this objective.
2. Explain why this solution is useless for downstream tasks.
3. Suggest **two** design changes used in modern SSL to avoid this collapse.

Solution: 1) Constant embeddings give a global minimum.

Suppose every image is mapped to the same constant vector c . Then for any two augmentations of the same image,

$$z^{(1)} = c, \quad z^{(2)} = c.$$

Therefore,

$$L = \|c - c\|_2^2 = 0.$$

Since squared Euclidean distance is always nonnegative, $L \geq 0$ for every possible model. So achieving $L = 0$ means this constant solution is a **global minimum**.

2) Why this is useless.

If every image has the same embedding, then the representation contains no information that distinguishes one image from another. A linear probe or downstream classifier cannot separate classes from identical features, so performance will be poor, often near chance level.

3) Possible anti-collapse design changes.

Examples include:

- use contrastive learning with negative samples (SimCLR-style),
- add variance/covariance regularization (e.g. VICReg-style),
- introduce teacher-student asymmetry with stop-gradient / centering / sharpening (BYOL/DINO-style).

Part D — Advanced Theoretical (2.0 points)

30. Why Scaled Dot-Product Attention Uses $1/\sqrt{d}$

In a transformer, attention scores are computed from a *query* vector $q \in \mathbb{R}^d$ and a *key* vector $k \in \mathbb{R}^d$ using their dot product

$$s = q^\top k = \sum_{j=1}^d q_j k_j.$$

In practice, transformers use the *scaled* score

$$\tilde{s} = \frac{q^\top k}{\sqrt{d}}.$$

To make the question self-contained, recall the following facts.

- The **expectation** $\mathbb{E}[X]$ is the average value of a random variable X .
- The **variance** $\text{Var}(X)$ measures how strongly X fluctuates around its mean:

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2].$$

- For any random variables $X_1, \dots, X_d$,

$$\mathbb{E}\left[\sum_{j=1}^d X_j\right] = \sum_{j=1}^d \mathbb{E}[X_j].$$

- If $X_1, \dots, X_d$ are independent, then

$$\text{Var}\left(\sum_{j=1}^d X_j\right) = \sum_{j=1}^d \text{Var}(X_j).$$

- For a constant a ,

$$\text{Var}(aX) = a^2 \text{Var}(X).$$

Assume that for every coordinate $j = 1, \dots, d$:

$$\mathbb{E}[q_j] = \mathbb{E}[k_j] = 0, \quad \text{Var}(q_j) = \text{Var}(k_j) = 1,$$

and all scalar random variables $q_1, \dots, q_d, k_1, \dots, k_d$ are mutually independent.

Tasks.

1. Show that for each coordinate,

$$\mathbb{E}[q_j k_j] = 0.$$

2. Show that for each coordinate,

$$\text{Var}(q_j k_j) = 1.$$

3. Using the previous results, compute

$$\mathbb{E}[q^\top k] \quad \text{and} \quad \text{Var}(q^\top k).$$

4. Compute

$$\text{Var}\left(\frac{q^\top k}{\sqrt{d}}\right).$$

5. Explain briefly why large unscaled dot products can make the softmax too sharp when d is large, and why dividing by $\sqrt{d}$ improves optimization stability.

Solution:**1) Expectation of one product.**

Since q_j and k_j are independent,

$$\mathbb{E}[q_j k_j] = \mathbb{E}[q_j] \mathbb{E}[k_j] = 0 \cdot 0 = 0.$$

2) Variance of one product.

Because the mean is zero,

$$\text{Var}(q_j k_j) = \mathbb{E}[(q_j k_j)^2].$$

Using independence,

$$\mathbb{E}[(q_j k_j)^2] = \mathbb{E}[q_j^2] \mathbb{E}[k_j^2].$$

Now

$$\mathbb{E}[q_j^2] = \text{Var}(q_j) + (\mathbb{E}[q_j])^2 = 1 + 0 = 1,$$

and similarly $\mathbb{E}[k_j^2] = 1$. Therefore

$$\text{Var}(q_j k_j) = 1 \cdot 1 = 1.$$

3) Mean and variance of the dot product.

Since

$$q^\top k = \sum_{j=1}^d q_j k_j,$$

linearity of expectation gives

$$\mathbb{E}[q^\top k] = \sum_{j=1}^d \mathbb{E}[q_j k_j] = \sum_{j=1}^d 0 = 0.$$

Also, the terms $q_j k_j$ are independent across coordinates, so

$$\text{Var}(q^\top k) = \sum_{j=1}^d \text{Var}(q_j k_j) = \sum_{j=1}^d 1 = d.$$

4) Variance after scaling.

Using $\text{Var}(aX) = a^2\text{Var}(X)$ with $a = 1/\sqrt{d}$,

$$\text{Var}\left(\frac{q^\top k}{\sqrt{d}}\right) = \frac{1}{d} \text{Var}(q^\top k) = \frac{1}{d} \cdot d = 1.$$

5) Why this matters for attention.

If we use the raw score $q^\top k$, then its variance grows with d . So for large feature dimension, attention scores tend to have larger magnitudes. After softmax, this can make the distribution too sharp: one token may receive almost all the weight, while many others get values near zero. That pushes softmax toward saturation and can make gradients weak or unstable.

Dividing by $\sqrt{d}$ keeps the variance of the score roughly constant (equal to 1 under these assumptions), so the softmax stays in a more reasonable regime and optimization becomes more stable.